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Portfolio replica of a target index

1. Introduction

Portfolio replication is one of the most relevant problems in quantitative finance, particularly when the objective is to reproduce the behaviour of complex benchmark indices through a reduced and tradable set of liquid instruments. Static allocation methods often fail to capture the evolving relationships among financial variables, especially during periods characterised by structural breaks, volatility clustering and regime shifts. Adaptive methodologies that account for time-varying dependencies are therefore essential to improve replication accuracy.
The present work addresses the replication of a synthetic benchmark, the *Monster Index*, through a set of tradable futures contracts spanning multiple asset classes. The analysis combines regression-based approaches, state-space methodologies and probabilistic regime-switching techniques to investigate whether dynamic portfolio construction outperforms static specifications in terms of tracking quality and economic feasibility. For each model, we additionally impose the UCITS constraints to ensure that the replication strategies remain economically realistic, risk-controlled, and implementable within institutional portfolio management standards. In particular, the portfolio must satisfy a one-month Value-at-Risk constraint,

$$\mathrm{VaR}_{0.99}^{1M} < 0.2,$$

or a maximum leverage constraint,

$$\max(\text{Leverage}) < 2.$$

Particular attention was devoted to preventing data leakage and look-ahead bias throughout the empirical framework. All rolling windows and estimation procedures were carefully designed to ensure strict temporal consistency between the information set available at time t and the subsequent portfolio decisions. In addition, the Markovian regime updates were implemented within a Bayesian filtering framework, ensuring that state probabilities were updated recursively using only information available up to the current observation, without incorporating future information into the estimation process.

2. Problem and Dataset

We address a *black-box portfolio replication* problem a target fund must be reconstructed using only liquid futures contracts, without knowing its actual holdings. The dataset consists of weekly Bloomberg observations from October 2007 to April 2021 (704 weeks), covering the Great Financial Crisis, the post-2009 low-volatility expansion, and the COVID-19 shock. Simple returns are computed as $r_t = (P_t - P_{t-1})/P_{t-1}$ for both the target indices and the investable futures.

3. Exploratory Data Analysis

The EDA confirms the standard stylised facts of financial returns. **QQ-plots** of futures and target returns display the typical signature of fat tails and excess kurtosis, providing clear evidence against the assumption of normality: returns align with the Gaussian quantile in the centre but deviate systematically at both ends. **Ljung–Box tests** on raw returns show weak autocorrelation for equity indices, near-white-noise behaviour for bonds and strong persistent autocorrelation for HFRXGL — consistent with hedge-fund return smoothing. The same test on **squared returns** rejects the null of no dependence at all lags and across all series, providing clear evidence of *volatility clustering*. The **ADF** and **Phillips–Perron** tests reject non-stationarity for all return series, supporting linear regression on returns despite the time-varying conditional variance. An important observation from the data was that the futures contract LLL1 became effectively illiquid ("dead") from a certain point onward. As this introduced distortions in the replication process, we excluded LLL1 from the final dataset to preserve model consistency and avoid contamination of the estimation sample.

4. Motivation for a Regime-Aware Framework

A single static replication model implicitly assumes that the relationship between target and futures remains constant over time; however, the EDA contradicts this assumption. Volatility clustering, fat tails and persistent autocorrelation in squared returns all indicate the presence of a non-stationary regime structure. Conditioning the replication on latent market states can therefore improve tracking stability and reduce estimation noise during structural transitions. For this reason we adopt a two-state Markovian framework: state 0 corresponds to quiet, low-volatility, trending market conditions, while state 1 captures volatile, high-volatility, and drawdown periods.

5. Failed Attempts

Before arriving at the final models, we explored several simpler configurations documented in the `failed_attempts` notebook. Starting from a plain weekly-rebalancing Elastic Net (OLS R1), we progressively added monthly rebalancing

(OLS R4), exponential sample weighting (EWM R4), and a long-only constraint (OLS R4 LO), then tested a model combination approach (Pooling MC) following O'Doherty, Savin Tiwari (2017) — "Hedge Fund Replication: A Model Combination Approach". Table 1 summarises the out-of-sample performance across all OLS configurations, while Table 2 presents the base versus regime-aware Kalman Filter results.

Table 1: Out-of-sample performance comparison across all regression-based models.

Config	IR	TEV (%)	Corr	Sharpe	Ret (%)	Vol (%)	MaxDD (%)	Avg GE	Avg TO	MSE
OLS R1	−0.081	3.80	0.862	0.88	5.68	6.47	12.7	1.47	10.06	2.77×10^{-5}
OLS R4	0.036	3.64	0.874	0.94	6.12	6.54	13.0	1.48	4.51	2.55×10^{-5}
EWM R4	0.126	3.79	0.863	0.99	6.47	6.52	11.9	1.78	5.36	2.75×10^{-5}
OLS R4 LO	−0.034	3.53	0.883	0.89	5.87	6.56	12.3	0.98	2.61	2.39×10^{-5}
Pooling MC	−0.153	3.76	0.883	0.73	4.91	6.74	11.6	0.48	0.69	2.72×10^{-5}
MS-EN	−0.035	3.94	0.872	0.78	5.36	6.85	10.8	1.55	5	2.96×10^{-5}

Table 2: Out-of-sample performance of the base and regime-aware Kalman Filter.

Model	IR	TEV (%)	Corr	Sharpe	Ret (%)	Vol (%)	MaxDD (%)	Avg GE	MSE
Kalman Base	−0.043	3.22	0.916	0.70	4.82	6.87	13.1	1.71	2.00×10^{-5}
Kalman Regime-Aware	0.098	3.22	0.916	0.77	5.76	7.51	11.8	1.40	2.00×10^{-5}

The two regime-aware models reported at the bottom of Table 1 — the Soft Markov Switching Elastic Net (MS-EN) and the regime-aware Kalman Filter — represent the final selected specifications. While the simpler configurations achieve competitive tracking metrics, none incorporates latent regime information; the regime-aware Kalman is the first model to achieve a consistently positive IR (0.098), and the MS-EN is the right trade-off between IR, max DD and computational costs. The following sections detail their construction and regime-aware design.

6. Hidden Markov Models and Regime Detection

Since regimes are not directly observable, we identify them through a Hidden Markov Model. Market behaviour is assumed to evolve according to an unobservable stochastic process governed by two latent states interpreted from a volatility perspective; regime probabilities are treated as continuous quantities rather than hard classifications, reflecting the gradual nature of regime transitions.

Latent states are estimated through the **Baum–Welch algorithm** on the initial training window. This expectation-maximisation procedure iteratively estimates the transition matrix and the state-conditional emission parameters, which are then used as informative priors for a sequential online filter. To preserve strict causality and avoid look-ahead bias, an **online Bayesian updating** framework based on Dirichlet–Normal-Gamma conjugate priors is layered on top of the Baum–Welch initialisation. Regime probabilities are updated sequentially as new observations become available, ensuring that future information is never implicitly incorporated into present estimations.

The estimated regime probabilities reveal economically meaningful patterns. A notable rise in the high-volatility probability emerges around 2011–2012, corresponding to the European sovereign debt crisis, with values peaking near 0.8 and reverting progressively toward stable conditions. A substantially stronger transition is detected during the 2020 COVID-19 shock, when the probability of the high-volatility state rapidly approaches unity. The framework therefore identifies major structural breaks consistently with well-known financial stress episodes while preserving causality. The resulting regime probabilities feed both the regression-based and state-space replication models described in the following sections.

7. Soft Markov Switching Elastic Net

The MS-EN combines three ingredients: (i) ElasticNet regression for portfolio weight estimation; (ii) a Baum–Welch HMM fitted on the rolling training window to estimate the transition matrix A, the regime means μ_s, and the regime volatilities σ_s; (iii) an online Bayesian forward filter that produces causal continuous probabilities $P(\text{high-vol} \mid t)$ for every observation in the window.

At each rebalancing step (every 4 weeks, with a 104-week rolling window chosen, because they give a better trade off between turnover cost and error), the loop builds two regime-conditional sample-weight vectors by combining the filtered probabilities with an adaptive exponential decay whose half-life is inferred from recent return activity. Two ElasticNet models are then fitted on the same data but with different sample weights: $\hat{w}_{\text{low}}$ uses $P(\text{low-vol} \mid t) \times$ decay with hyperparameter grid `GRID_LOW_VOL` (full range), while $\hat{w}_{\text{high}}$ uses $P(\text{high-vol} \mid t) \times$ decay with a restricted grid favouring sparser, more regularised solutions. The two regime-conditional weight vectors are combined through **soft blending**:

$$w_{\text{MS}} = p_{\text{low}}\, \hat{w}_{\text{low}} + (1 - p_{\text{low}})\, \hat{w}_{\text{high}},$$

where p_{low} is the 4-week mean of the filtered probability. Two weekly risk controls are applied: a turnover gate that suppresses small rebalances, and a VaR-based scaling that caps the 4-week 99% portfolio VaR at 20% through a cascade of estimators, depending on how much observations we have (filtered bootstrap → GARCH-t → static Student-t). When regime separation in the training window is insufficient (fewer than 20 observations with majority probability per regime), the model degrades gracefully to a plain ElasticNet with time-decay sample weights only. The metrics on the test set are reported in Table 3.

8. Kalman Filter

The **Kalman Filter** treats portfolio weights as a latent state vector evolving according to a linear state-space model, updated recursively at each observation through a predict–update cycle without requiring a fixed training window. The process noise covariance Q_t is regime-dependent: during turbulent periods, larger Q_t enables faster weight adaptation; during stable environments, smaller Q_t favours smoother transitions and reduced turnover.

The regime-aware Kalman achieves the strongest replication performance, with a correlation of 0.916, a tracking error volatility of 3.22%, and an MSE of 2×10^{-5}. Transaction costs average approximately 13.8 basis points per rebalance, confirming operational feasibility. The MS-EN, while marginally weaker in raw tracking metrics, exhibits the lowest maximum drawdown (10.8%) and a structurally different error profile driven by regime-conditional regularisation. These complementary properties motivate the construction of an ensemble combining both models.

9. Stacking Methodology and Final Statistical Analysis

To further improve replication quality, an ensemble methodology based on **Soft Bayesian Stacking** is introduced. The rationale is that the MS-EN and the regime-aware Kalman capture distinct dimensions of the financial dynamics — the former exploits regime-conditional regularised regression, the latter sequential state-space updating — and their aggregation may produce more robust replication outcomes through diversification across modelling assumptions.

The stacking framework combines the predictions generated by the two models through posterior weights that evolve over time. At each step, the posterior weight of each model is updated using its recent predictive log-likelihood, with a forgetting factor $\lambda = 0.97$ to discount older evidence. The combined prediction is the convex combination of the individual predictions, with weights bounded in $[0.15, 0.85]$ to prevent collapse to a single model.

Table 3: Extended out-of-sample performance comparison across standalone models and the Soft Bayesian Stack (common OOS period: 599 observations, 2009-10-27 to 2021-04-13).

	Replication quality					Risk-adjusted				Tail risk (weekly)				Operational	
Model	**Corr**	**TEV (%)**	**IR**	**RMSE (%)**	**Hit (%)**	**Sharpe**	**Sortino**	**Calmar**	**MaxDD (%)**	**VaR$_{99}$ (%)**	**CVaR$_{99}$ (%)**	**Skew**	**Kurt**	**Ann.Ret (%)**	**Ann.Vol (%)**
MS-EN (Dir-NG)	0.872	3.94	−0.035	0.544	86.5	0.78	0.992	0.495	10.8	−2.673	−3.569	−0.520	3.22	5.36	6.85
Kalman regime-aware	0.916	3.22	0.098	0.447	87.6	0.77	0.965	0.488	11.8	−2.844	−3.959	−0.599	4.05	5.76	7.51
Stacking	**0.914**	**3.27**	**0.055**	**0.453**	**88.5**	**0.78**	**0.983**	0.482	11.7	**−2.748**	**−3.827**	−0.623	4.04	5.62	7.22
Target (Monster Index)	—	—	—	—	—	0.68	0.837	0.345	15.9	−2.783	−4.364	−1.045	7.55	5.50	8.04

The Soft Bayesian Stack dominates both standalone models on the core replication metrics: correlation reaches 0.914, tracking error volatility narrows to 3.27%, the Information Ratio turns positive at 0.055, and the weekly hit rate of 88.5% is the highest across all models. Tail risk is also reduced, with a monthly Cornish–Fisher VaR of 5.04% against the target's 7.26%, confirming UCITS compliance. The average Bayesian weight of 0.789 assigned to the Kalman reflects its stronger performance in the low-volatility 2012–2019 window, while the MS-EN contributes more during stress episodes, confirming that the two models carry complementary information that justifies their combination.

Model selection rationale

The MS-EN is selected for its regime-conditional drawdown control and its structurally different error profile relative to the Kalman: bounded rolling window versus cumulative state-space recursion, regime-conditional regularisation versus sequential Bayesian updating. This methodological asymmetry produces complementary residuals that Soft Bayesian Stacking exploits dynamically, concentrating weight on the locally more accurate model rather than applying a fixed combination. The result is a replication strategy that is more robust than either component in isolation across the full range of market conditions observed in the sample. It should be noted that model selection was performed over the full OOS window, introducing a mild look-ahead bias that should be kept in mind when interpreting the reported metrics.

10. Risk Management Analysis: VaR and Stress Testing

Beyond replication accuracy, model evaluation requires an assessment of downside risk and resilience under adverse market conditions. Risk is measured through **Value-at-Risk** (VaR) and **Expected Shortfall** (ES) at both backtesting and regulatory frequencies.

Backtesting VaR is computed weekly through a cascade of estimators: a filtered historical simulation, a parametric GARCH-t specification, and a static Student-t approximation as the final fallback. The regulatory measure aggregates returns to monthly frequency and computes a 99% VaR through a Cornish–Fisher expansion accounting for skewness and excess kurtosis. Expected Shortfall complements VaR by capturing the magnitude of losses beyond the confidence threshold.

The Kupiec proportion-of-failures test on weekly VaR exceptions does not reject the null of correct unconditional coverage (p-value > 0.20), confirming that the estimates are statistically consistent with realised losses over the out-of-sample period.

Stress testing complements the backtesting analysis by evaluating the portfolio under scenarios calibrated to historical episodes of severe market dislocation (GFC 2008, European debt crisis 2011–2012, COVID-19 shock 2020). In each scenario, simulated maximum monthly losses remain below the regulatory VaR estimate, supporting the strategy's viability under adverse conditions. The risk framework reinforces the conclusions of the replication analysis: the stacked strategy achieves robust tracking with controlled tail risk, disciplined leverage and stable turnover.